FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE

JEE(Advanced)-2015

ANSWERS, HINTS & SOLUTIONS

CRT –II
(Paper-1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	B	A	A
2.	A	D	A
3.	B	D	B
4.	D	A	D
5.	A	C	C
6.	A	A	C
7.	B	A	D
8.	C	A	C
9.	C	A	D
10.	A	C	D
11.	A,B	B, D	A, C
12.	B,D	B, C	C, D
13.	C,D	A, B, C, D	A, D
14.	A,B,C	A, B, C	A, C, D
15.	A,B,D	C, D	A, B
1.	2	1	4
2.	6	3	0
3.	5	2	4
4.	2	1	6
5.	6	3	9

Physics

PART – I

SECTION – A

1. Optical path length of ray entering through S_1

$$x_1 = \int_0^L \mu dx = L + \frac{aL^2}{2}$$

Path difference of ray from S_1 and $S_2 = \Delta x = \frac{aL^2}{2}$

For minima at O

$$\frac{aL^2}{2} = \frac{\lambda}{2}$$

2. Fundamental frequency $f_0 = \frac{v}{2L}$

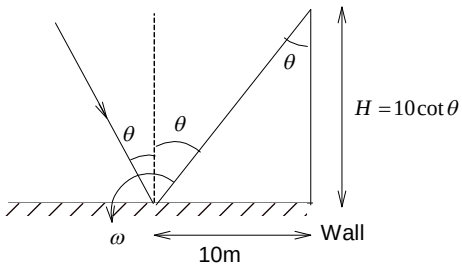
Beat frequency = $f_1 - f_2$

$$= \frac{v}{2\left(\frac{L}{2} - \Delta L\right)} - \frac{v}{2\left(\frac{L}{2} + \Delta L\right)}$$

$$= \frac{v}{2}\left[\frac{2}{(L - 2\Delta L)} - \frac{2}{(L + 2\Delta L)}\right]$$

$$= 2f_0 L\left[\frac{4\Delta L}{L^2 - 4\Delta L^2}\right] \approx \frac{8f_0 \Delta L}{l}$$

3. When mirror rotates, by ω, reflected ray rotates by $2\omega = 36$ rad/s.



speed of spot = $\frac{dH}{dt} = \frac{d[10\cot\theta]}{dt}$

$$= \left[-10\,\text{cosec}^2\theta \frac{d\theta}{dt}\right]$$

$$= \left[\frac{-10}{(0.6)^2} \times 36\right]$$

= 1000 m/s

4. Energy of nth state in H-atom is same as energy if 3nth state in Li^{++}.

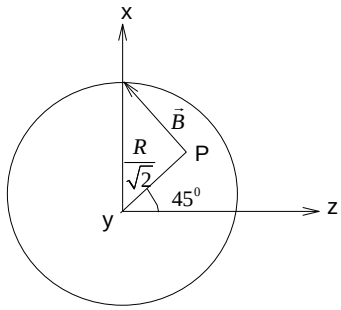
5. $v = 2f_0(l_2 - l_1)$

$$\left(\frac{\Delta v}{v}\right)_{max} = \frac{\Delta f_0}{f_0} + \frac{\Delta l_2 + \Delta l_1}{l_2 - l_1}$$

$$= \frac{1}{100} + \frac{0.1 + 0.1}{74 - 24}$$

$$= 1.2\%$$

6. Magnitude of field at Point $P\left(\frac{R}{2}, y, \frac{R}{2}\right)$ is



$$B = \frac{\mu_0 Ir}{2} = \frac{\mu_0 I}{2\pi R^2} \times \frac{R}{\sqrt{2}} = \frac{\mu_0 I}{2\sqrt{2}\pi R}$$

Unit vector in the direction of field, $\hat{B} = \frac{\hat{i} - \hat{k}}{\sqrt{2}}$

7. $i = \frac{10}{10} = 1A$; $V = iR = 1 \times 4 = 4V$

8. $\left(2 - \frac{6}{8} \times 2\right) = \left(\frac{12}{8+16}\right) 4x \Rightarrow x = 25$ cm.

9. Ammeter and 0.9 Ω is in series
$10 \times 10^{-3}(9.9) = i \times 0.1 = i = 0.99$
So current = 0.99 + 0.01 = 1 Amp.

Chemistry

PART – II

SECTION – A

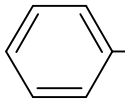
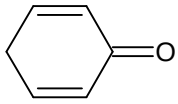
1. VBT and MOT both consider wave nature of e^-.

2. H_3PO_4 can act as a dehydrating agent.

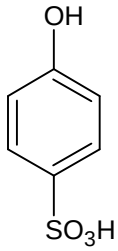
4. Hydrocarbon part is hydrophobic part

5. $\left[Cr\left(H_2O\right)_6\right]^{3+}$ has maximum CFSE

6. $\upsilon = \frac{v}{2\pi r}$ and both v and r can be determined by Bohr's theory .

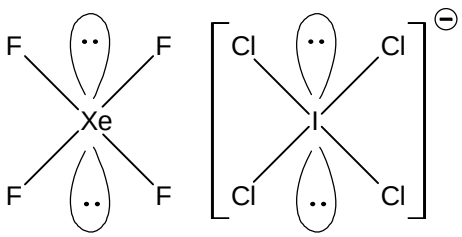
7.  is aromatic hence ΔG_f of  is maximum in acidic medium.

8. $RX \xrightarrow{LiAlH_4} Alkene$

9.  is major at high T, at high temperature para hydrogen convert into ortho hydrogen.

10. $\left[Cu\left(NH_3\right)_4\right]^{2+}$ has square planar shape

12. (C) : $[Cr(NH_3)_5Cl]NO_2$
: $[Cr(NH_3)_5NO_2]Cl$
: $[Cr(NH_3)_5ONO]Cl$

15. 

SECTION –C

1. In H_2O and CH_3CH_2OH mixing shows +ve deviation while in other cases it shows –ve deviation.

2. Electron will reach n = 4, so the maximum possible transitions for single electron are $4 \rightarrow 3$, $3 \rightarrow 2$, $2 \rightarrow 1$.

3. $nM \longrightarrow M^{n+}_{n_A} + ne^- \quad (\text{at anode})$

$M^{n+}_{n_c} + ne^- \longrightarrow 2M \quad (\text{at cathode})$

Net all reaction: $M^{n+}_{n_c} \longrightarrow M^{n+}_{n_A}$

$$\therefore 0.0295 = \frac{0.059}{n}\log\frac{\left[M^{n+}_{n_C}\right]}{\left[M^{n+}_{n_A}\right]}$$

$\therefore n = 2$

4. $\left[Pt(en)_3\right]^{4+}$ is optically active but does not show geometrical isomerism.

5. $\underset{(H_2O_2)}{N_1V_1} = \underset{KMnO_4/H^+}{N_2V_2}$

$2 \times 25 = N \times 25$

$N = 2$

Again $\underset{KMnO_4/H^+}{N_1V_1}$ = gm eq. of (X)

$\therefore$ 2×1 = gm eq. of (X)

and n factor of x = 2

Since X is converted into single e species Hence X is Li^{2+} and light radiation of 114 $\overset{o}{A}$ corresponds to 108.77 eV.

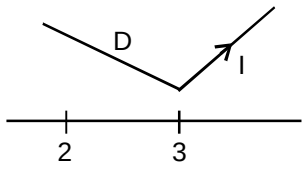
$\therefore$ e is excited into 3rd energy level and total 3 transitions are possible.

Mathematics

PART – III

SECTION – A

1. $f(x) = x^{\frac{3}{2}} + x^{\frac{-3}{2}} - 4\left(x + \frac{1}{x}\right)$



$f(x) = \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^3 - 3\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right) - 4\left[\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2 - 2\right]$

Let $\sqrt{x} + \frac{1}{\sqrt{x}} = t$ $(x > 0)$

Let $g(t) = t^3 - 3t - 4t^2 + 8$

Now $g(t) = t^3 - 4t^2 - 3t + 8$ where $t \in [2, \infty)$

$g'(t) = 3t^2 - 8t - 3 = (t - 3)(3t + 1)$; $g'(t) = 0 \Rightarrow t = 3$ $(t \neq -1/3)$

$g''(t) = 6t - 8$

$g''(3) = 10 > 0 \Rightarrow g(3)$ is minimum

$g(3) = 27 - 9 - 36 + 8 = -10$

2. Substituting $\ln x = t$, we get $\int_1^{2010} \left(1 + \frac{1-t}{t(t-(\ln t))}\right) dt = 2009 + \int_1^{2010-\ln 2010} \frac{-1}{u} du$ $(u = t - \ln t)$

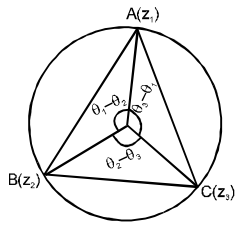
$= 2009 - \ln(2010 - \ln 2010)$

3. Since, $|z_1| = |z_2| = |z_3| = 1$ and $z_1 + z_2 + z_3 = 0$

$\therefore$ Triangle ABC is an equilateral triangle

$\therefore \theta_1 - \theta_2 = \theta_2 - \theta_3 = \theta_3 - \theta_1 = \frac{2\pi}{3}$

$\therefore \cos(\theta_1 - \theta_2) + \cos(\theta_2 - \theta_3) + \cos(\theta_3 - \theta_1) = -\frac{3}{2}$



4. In $(0, 2\pi)$ check for 1, 2, 3, 4, 5, 6 function is not continuous for 4

Hence, (x, y) is (4, –1)

5 $2x^3y\,dy + (1 - y^2)(x^2y^2 + y^2 - 1)\,dx = 0$, Let $1 - y^2 = t \Rightarrow 2y\,dy = -dt$

$-x^3\,dt + t(x^2(1 - t) - t)\,dx = 0 \Rightarrow -x^3dt + tx^2dx - t^2x^2dx - t^2dx = 0$

$x^2(t\,dx - x\,dt) = t^2(x^2 + 1)\,dx \Rightarrow \frac{tdx - xdt}{t^2} = \left(\frac{x^2+1}{x^2}\right)dx \Rightarrow d\left(\frac{x}{t}\right) = \left(1 + \frac{1}{x^2}\right)dx$

Integrate $\frac{x}{t} = x - \frac{1}{x} + c \Rightarrow x = \frac{x^2-1}{x}(1-y^2) + c(1-y^2)$

$x^2 = (x^2 - 1)(1 - y^2) + xc(1 - y^2) \Rightarrow 0 = -x^2y^2 - 1 + y^2 + xc - xcy^2$

$\Rightarrow x^2y^2 = (cx - 1)(1 - y^2)$

6. Total number of coordinates in $S = 3 \times 4 \times 5 = 60$

$\therefore$ Total no. of way of selecting two points $= {}^{60}C_2$

Now the selection of (x_1, y_1, z_1) and (x_2, y_2, z_2) should be in such a way that

$x_1 + x_2$, $y_1 + y_2$ and $z_1 + z_2$ are even

Now x_1, x_2, may be

$\therefore$ Favourable points $= 5 \times 8 \times 13 = 520$

$\therefore$ Probability $= \frac{520}{30 \times 59} = \frac{52}{177}$

7. Number of ways = total – those in which at least three boys between the girls = $\underline{|17} - a$

Now, x G_1 y G_2 z

x + y + z = 15

Min x = 0 max x = 12

Min y = 3 max y = 15

Min z = 0 max z = 12

Total solution = coefficient of x^{15} in $(x^0 + \ldots. x^{12})(x^3 + x^4 + \ldots. + x^{15})(x^0 + \ldots. + x^{12})$

$= \ldots. x^{12}$ in $(1 + x + \ldots. x^{12})^3 = \ldots. x^{12}$ in $\left(\frac{1-x^{13}}{1-x}\right)^3 = \ldots. x^{12}$ in $(1-x^{13})^3(1-x)^{-3}$

$= {}^{12+3-1}C_2 = {}^{14}C_2$

$\therefore\ a = {}^{14}C_2 \times \underline{|2} \times \underline{|15}$ ways $= \underline{|17} - 91 \times \underline{|2}\ \underline{|15}$

8. Let $\vec{d}$ makes angle α from $\vec{a}$, $\vec{b}$ and $\vec{c}$ then $\cos^2\alpha + \cos^2\alpha + \cos^2\alpha = 1$

($\because$ $\vec{a}$, $\vec{b}$, $\vec{c}$ mutually perpendicular)

$\Rightarrow \cos\alpha = \pm\frac{1}{\sqrt{3}}$

$|\vec{a}+\vec{b}+\vec{c}+\vec{d}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + |\vec{d}|^2 + 2(\vec{a}\cdot\vec{b} + \vec{a}\cdot\vec{c} + \vec{a}\cdot\vec{d} + \vec{b}\cdot\vec{c} + \vec{b}\cdot\vec{d} + \vec{c}\cdot\vec{d})$

$= 1 + 1 + 1 + 1 + 2\,(0 + 0 + \cos\alpha + 0 + \cos\alpha + \cos\alpha) = 4 + 6\cos\alpha = 4 \pm 2\sqrt{3}$

9. The line makes equal angle with planes hence with normal

i.e. $\frac{\vec{a}\cdot\vec{n}_1}{|\vec{a}||\vec{n}_1|} = \frac{\vec{a}\cdot\vec{n}_2}{|\vec{a}||\vec{n}_2|}$

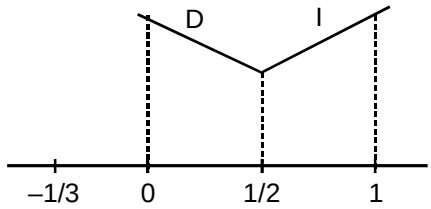
$\vec{a} = k\hat{i} + 2\hat{j} - 12\hat{k},\ \vec{n}_1 = 2\hat{i} + \hat{j} + 3\hat{k},\ \vec{n}_2 = \hat{i} + 2\hat{j} - 3\hat{k}$

$\frac{2k + 2 - 36}{\sqrt{14}} = \frac{k + 4 + 36}{\sqrt{14}} \Rightarrow k = 74$

10. $f'(x) = 12x^2 - 2x - 2$

$= 2[6x^2 - x - 1]$

$= 2(3x + 1)(2x - 1)$



11. S_1: $x^2 + y^2 - 4x - 81 = 0$

$S_1 - S_2 = 0$

x = 0 equation of AB

Now put A in S_1 or S_2

a = ± 9

Now PA, $y - 9 = mx \Rightarrow y = mx + 9$

Now let $P(x_1, y_1)$ and $Q(x_2, y_2)$

S_2: $x^2 + y^2 + 24x - 81 = 0$

28x = 0

Now let A ≡ (0, a) $\therefore$ A is on x = 0

$0 + a^2 + 0 - 81 = 0$

A = (0, 9), B = (0, –9)

$\Rightarrow$ BQ, $y + 9 = mx \Rightarrow y = mx - 9$

Now solve S_1 and PA then $x_1 = \frac{-18m - 24}{m^2 + 1}$ and $y_1 = \frac{-9m^2 - 24m + 9}{m^2 + 1}$

Now solve S_2 and BQ then $x_2 = \frac{18m + 4}{m^2 + 1}$ and $y_2 = \frac{9m^2 + 4m - 9}{m^2 + 1}$

Let mid pt of PQ is (h, k) then $x_1 + x_2 = 2h$ and $y_1 + y_2 = 2k$

$x_1 + x_2 = \frac{-20}{m^2 + 1} = 2h$

$\frac{h}{k} = \frac{1}{m}$

$y_1 + y_2 = \frac{-20m}{m^2+1} = 2k \Rightarrow \frac{-20}{\left(\frac{k}{h}\right)^2 + 1} = 2h$

$h(h^2 + k^2 + 10h) = 0$

12. For one local max and min $f'(x) = 0$ will have two real and unequal roots
$f'(x) = 6x^2 - 6(2 + \lambda)x + 12\lambda \Rightarrow D = (2 + \lambda)^2 - 8\lambda > 0 \Rightarrow (\lambda - 2)^2 > 0 \Rightarrow \lambda \in R - \{2\}$

13 P lies on the line $2x + 3y + 1 = 0$ ..... (1)
$|PA - PB|$ is maximum if P, A, B are collinear.
Equation of the line AB is $x + y = 2$ ..... (2)
P is the point of intersection of (1) and (2) which is $(7, -5)$
Equation of perpendicular bisector of AB is
$y = x$ ..... (3)
For minimum value of $|PA - PB|$, P is the point of intersection of (1) and (3) which is $\left(-\frac{1}{5}, -\frac{1}{5}\right)$

14. Since $x^2 - 3x + 2$ is factors of $x^4 - px^2 + q$, then roots of $x^2 - 3x + 2 = 0$
i.e. 2, 1 are also roots of $x^4 - px^2 + q = 0$
but if a is roots of $x^4 - px^2 + q = 0$, then $-a$ is also
roots of $x^4 - px^2 + q = 0$ are $\pm 1, \pm 2$
and $1 - p + q = 0$ and $16 - 4p + q = 0$, $p = 5$ and $q = 4$

15. $f(x) = ae^{2x} + be^x + cx$
Since $f(0) = a + b$ i.e. $a + b = -1$ ..... (1)
$f'(x) = 2ae^{2x} + be^x + c$
$\therefore f'(\log 2) = 2ae^{2\log 2} + be^{\log 2} + c$
$= 8a + 2b + c = 8a + 2b + c = 31$ ..... (2)

$\int_0^{\ell n 4} (ae^{2x} + e^x + cx - cx)\,dx = \int_0^{\ell n 4} (ae^{2x} + be^x)\,dx$

$= \left(\frac{ae^{2x}}{2} + be^x\right)\Big|_0^{\log 4} = \frac{a\,e^{2\ell n 4}}{2} + be^{\ell n 4} - \frac{a}{2} - b = 8a + 4b - \frac{a}{2} - b = \frac{15a}{2} + 3b = \frac{39}{2}$

i.e. $15a + 6b = 39$ ..... (3)
From (1), (2) and (3) we get $9a = 45$
$\therefore a = 5, b = -6$ and $c = 3$

SECTION – C

1. Straight line from Origin is $y = mx$

$4x + 2y = 3$, $y = mx$, $x_1 = \frac{3}{(2m+4)}$, $y_1 = \frac{3m}{(2m+4)}$

$2x + y + 6 = 0$, $y = mx$, $x_2 = \frac{-6}{(m+2)}$, $y_2 = \frac{-6m}{(m+2)}$

$O = \frac{\frac{-6m}{(m+2)} + \frac{3mk}{2(m+2)}}{k+1} \Rightarrow O = \frac{\frac{-6}{m+2} + \frac{3k}{2(m+2)}}{k+1} = \frac{-6 + \frac{3k}{2}}{k+1} = 0 \Rightarrow \frac{3}{2}k = 6,\ k = \frac{12}{3} = 4$

2. $\cos^{-1} x + \cos^{-1} 2x = -\pi$, $\cos^{-1} x \in [0, \pi]$
$\cos^{-1} x + \cos^{-1} 2x = -\pi \Rightarrow$ no solution

3. abc = 1, trace A = a + b + c > 0

$$I = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}\begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix} = \begin{bmatrix} a^2+b^2+c^2 & ab+bc+ca & ac+ab+bc \\ ab+bc+ca & b^2+c^2+a^2 & bc+ca+ab \\ ca+ab+bc & bc+ca+ab & c^2+a^2+b^2 \end{bmatrix}$$

$\therefore a^2 + b^2 + c^2 = 1$, $ab + bc + ca = 0$
$(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca) = 1$
$\therefore a + b + c = 1$
$\therefore a^3 + b^3 + c^3 = 3abc + (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = 3 + 1(1 - 0) = 4$

4. $f^3(x) = \int_0^x t \cdot f^2(t)dt$ ..... (1)

Differentiate with respect to x

$3f^2(x) \cdot f'(x) = x \cdot f^2(x) \Rightarrow f'(x) = \frac{x}{3}$

$\Rightarrow f(x) = \frac{x^2}{6} + C$ ..... (2)

But from (1) $f^3(0) = 0 \Rightarrow f(0) = 0$

$\therefore$ From (2) C = 0 $\therefore f(x) = \frac{x^2}{6} \Rightarrow f(6) = \frac{6^2}{6} = 6$

5. The line $x - 3\sqrt{5}y = 0$ and $\sqrt{5}x - 2y = 13$ intersect at the point. Since the given hyperbola passes through this point,

$\frac{(3\sqrt{5})^2}{a^2}\frac{1}{b^2} = 1$ or $\frac{45}{a^2} - \frac{1}{b^2} = 1$ ..... (1)

Also length of the latus rectum = $\frac{2b^2}{a} = \frac{4}{3}$ (given) ..... (2)

From (1) and (2) we get $\frac{45}{a^2} - \frac{3}{2a} = 1$

$\Rightarrow a = 6$ or $-\frac{15}{2}$

$\Rightarrow a > 0$ therefore we have $a = 6$
$\Rightarrow b^2 = 4$
$\therefore e^2 = \frac{10}{9}$ hence $9e^2 - 1 = 10 - 1 = 9$